

Energy dissipation and behavioural regimes in an autonomous double pendulum subjected to viscous and friction damping

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Abstract

Nonlinear dynamics of a system of a compound double pendulum is experimentally analyzed and numerically verified. The experiments are carried out for the combination of initial angles in the range $0 \leq \theta_0, \phi_0 \leq 180^\circ$, with zero initial angular velocity and with an increment of 1° . The motion of the pendulums are captured in the video and the individual pendulums are tracked by using image processing. The energy dissipation in the system is evaluated by using the Rayleigh dissipation function to include non-conservative forces as well, as a lumped parameter in the Lagrangian equation. Furthermore, the experimental and the numerical results are brought to agreement by tuning the dissipation factors. The linear, nonlinear and chaotic behaviour of the system is classified in the coordinate plane. The behavioural regime represents the dependency of the nature of the system on the total energy and the difference in instantaneous amplitudes of the pendulums. Uncertainty in the behaviour of the system is observed in the transition region from the nonlinear to the chaotic regime due to the sensitivity of the system to the initial conditions. The dependence of the dissipation factors on the angular displacements are represented as a surface plot which highlights the dominance of viscosity in the higher amplitude oscillations and the dominance of contact friction in the lower amplitude oscillations. This phenomenon is established by direct comparison.

Keywords: double pendulum, Rayleigh dissipation function, dissipation factors, viscous damping, frictional damping

1. Introduction

The sensitivity of a system governed by deterministic laws [1] to minuscule changes in the initial conditions, leads to its unpredictable and uncertain behaviour. This behaviour is generally termed as chaos which can be seen in phase plane and phase space analysis. Prediction of the behaviour of a system under chaos is extremely difficult, while at the same time, experimental measurements of the responses of such systems are equally laborious and cumbersome, sometimes resulting in erroneous data.

Additionally, the difficulty is encountered because of the unattainable level of precision in the measurement of the initial conditions [2]. The simplest of the mechanical systems that have been reported to exhibit chaotic behaviour is a double pendulum system. The analysis of the motion of a planar compound double pendulum for different initial conditions open up a paradigm to understand the linear, the nonlinear and the chaotic dynamics; and hence by extension the behavioural regimes.

Closed-form solutions are available for the single undamped pendulum [3, 4] and the damped pendulum [5, 6] and it has been also found for the periodic and the quasiperiodic motion of the double pendulum [7]. Several numerical studies have been documented on the behaviour of the single pendulum [8], the double pendulum [9, 10, 11] and the triple pendulum [12] systems. Double pendulum exhibits chaotic behaviour in a four-dimensional phase space, which has been experimentally reported for a system of compound double pendulums with a variable centre of gravity [13], slender bars [14, 15] and asymmetrically oriented square plates [16] as its elements. The analysis of the compound double pendulum aids in the designing and understanding of multibody physical systems [17, 18] and thus carry considerable importance.

Fourier series analysis of a simple pendulum is used [19] in the determination of transition from the linear to the non-linear oscillations. In previous works, the response plot [20], the phase portrait [21], the Poincaré map [16], the power spectrum [22], the Lyapunov exponent [23] and the bifurcation diagram [9] are used to detect chaos in dynamical systems. Modern approaches [24] of First Lyapunov Indicator (FLI), Dynamic Lyapunov Indicator (DLI) and Smaller Alignment Index (SALI) has been verified with the classical approaches to determine the chaotic behaviour. The transition from the nonlinear regime to the chaotic regime is experimentally determined [25] and the dependency of the behaviour on the energy as well as the orientation of the system is well documented.

Various methodologies are used for tracking the pendulum, which includes the usage of an accelerometer [26], a stroboscope [14], an optical encoder [23], a V-scope microcomputer [20], a combination of magnet and coils [27], a combination of flashgun and fluorescent screen [28], a sonic rangefinder [29] and video processing by tracking the markers [16, 30, 25] using opensource software or pre-built tools. It is observed that the most preferred methodology are the once which are of non-contact mode for the obvious reason of the possible fluctuation in the experimental results due to the intervention of the sensory equipment.

In multibody systems the main contributors for the friction due to contact forces [31] are pure dry sliding friction, stick-slip effect, viscous friction due to lubricants, Stribeck effect [32] and frictional lag. In addition, the viscous damping due to the medium [33] also contribute to the energy dissipation. Collectively these losses are dissipated in the form of heat and sound energy. Several experimental works have been carried out on the single pendulum [34, 29, 35] accounting for these damping forces. Rayleigh dissipation function can be used in the solution to mechanical systems involving the friction depending linearly and nonlinearly on the velocity, including the Coulomb

friction[36]. Thus, Rayleigh dissipation function is adopted to account for both the linear damping and the nonlinear damping. The energy dissipated is lumped together by considering the Rayleigh damping constant as a fitting parameter, which is found by comparing the analytical results with the experimental results. Rayleigh dissipation function has been used as a fitting parameter in modelling various multibody systems, both numerically [37] and experimentally [30, 38].

From the available literature, it can be stated that energy dissipation in a double pendulum system is not thoroughly addressed and examined. This has led to the formulation of this work, where a comprehensive experimental study taking into account the energy dissipation is presented. Rayleigh dissipation function is included in the mathematical formulation and dissipation factors are used as fitting parameters. In this paper, the behaviour of the system is experimentally investigated and numerically verified for the entire range of combinations of the primary and the secondary angles in the step of 1° , with an operational range of $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$ for 32760 combinations. The transition from the linear regime to the nonlinear regime is decided when the deviation of the simulated results from the results of the linearized equation of motion is within 1% of their trajectory, while the transition from the nonlinear regime to the chaotic regime is determined by using the phase portraits and the Poincaré sections.

A planar compound double pendulum is chosen to determine different behavioural regimes of a chaotic deterministic system because of its rich chaotic behaviour, the simplicity in its replication into a physical model, lesser rate of energy loss, ease of tracking and solvable mathematical model. The upper and the lower pendulums are anodized with white and black colour respectively for better tracking of the pendulums. For the scope of our experiment, a non-contact mode of measurement is imperative and hence video processing technique is chosen for its minimal and non-additive error in measurement. The pendulums are detected as a complete unit based on their individual colours and their locations are determined with reference to their individual hinge positions. This technique further improves the accuracy of measurement by diminishing the effects of warping of the pendulum in comparison to the cited video processing methods which uses markers as a reference while tracking.

The equations of motion of the system are derived using Lagrangian formulation. As a general closed-form solution is unattainable, numerical solutions are used for analysis. When the initial angles are small, the double pendulum behaves similar to a system of two spring coupled oscillators and as the initial angular displacement(s) increases, the response of the system gradually transitions to the nonlinear behaviour and then subsequently to the chaotic behaviour at increased energy levels. This transition is presented in the right quadrant of the combination of initial angles. A surface equation is fit to the Rayleigh dissipation factors to generalize the relationship between the said dissipation factors and the parameters of the system.

In the further organization of the paper, the experimental setup and methodology of measurement are elaborated, followed by the governing equations of motion and the

scheme of solution. Furthermore, experimental and numerical results are assessed in detail followed by the discussions on the analysis of errors in measurement, uncertainty in the classification of regimes, dominance of viscous damping and friction damping at different stages of pendulum oscillation is explained. Finally, a note on the effects of the amplitude difference (ψ) on the duration of oscillation is presented after clearly identifying the amplitudes at which viscous and friction effects are predominant.

2. Experimental Model and Methods

The experimental setup is shown in Figure 1(a) consists of a compound double pendulum made from aluminium, which is firmly held by an inverted L shaped frame and fixed to a rectangular base. The assembly consists of a primary (upper) pendulum made of two parallel bars, supporting a secondary (lower) pendulum made of a single bar. Two deep groove ball bearings are press-fitted to the individual bars of the primary pendulum, through which a stud of diameter 8 mm is press-fitted which is further secured to the L-frame using a fastener. The secondary pendulum is coupled to the primary pendulum by a stud of 8 mm diameter, which runs by the axis of rotation of the bearing of the secondary pendulum and through the lower holes of the primary pendulum. The axis of rotation for both the pendulums lies at a distance of 20 mm from their respective free ends.

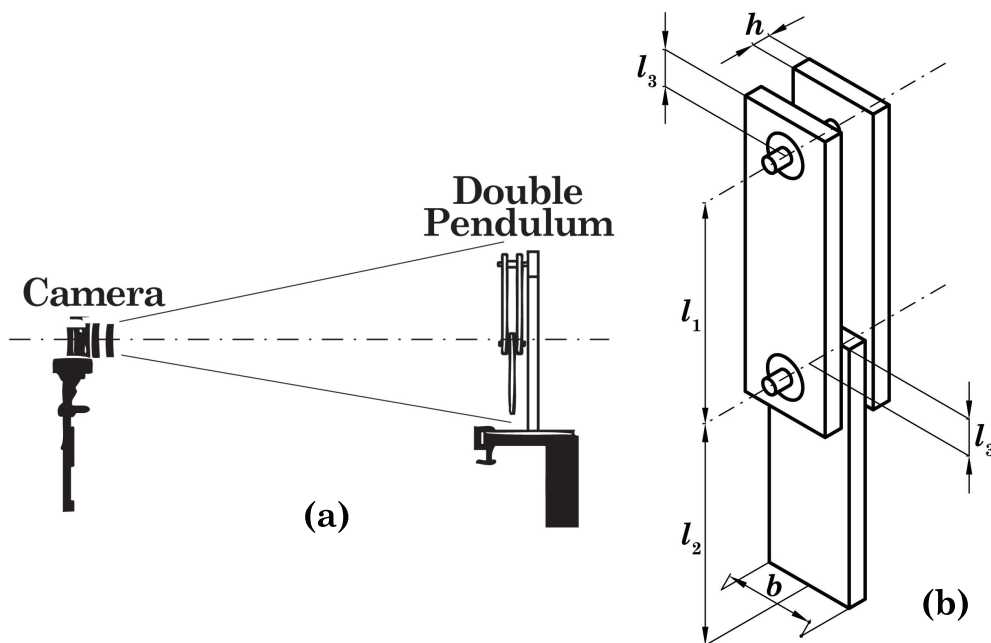


Figure 1. Experimental setup and model (a) Schematic of the apparatus (b) Geometric model.

The deep groove ball bearings used in the setup are made of stainless steel with an outer diameter of 22 mm, an inner diameter of 8 mm and a thickness of 7 mm. The

balls used are of diameter 4 mm, which are rated to withstand a static load of 1.37 kN and a dynamic load of 3.45 kN [39]. The limiting speed of the bearings are rated at 48000 rpm. For steel and steel material contact, the rolling friction coefficient is rated as 0.001–0.002 and for the clean and dry conditions the static friction coefficient is rated as 0.5–0.8 and the dynamic friction coefficient is rated as 0.42.

The pendulums are free to rotate in a two-dimensional plane about the axis along with the studs. Angular measurements are laser engraved in a circular fashion on the surface of the pendulum about its axis of rotation, along with a fixed pointer pointing the instantaneous angle. The operational range(s) of the pendulum(s) are $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$. The distance between the two hinges in the primary pendulum(l_1), the distance from the second hinge to the free end of the secondary pendulum(l_2), the shortest distance from the hinges to the respective free ends of the pendulums(l_3), the width of the pendulums(b), the thickness of the individual bars of the pendulums(h), collective rotary mass of the primary pendulum(m_1), the collective rotary mass of the secondary pendulum(m_2) are listed in Table 1 and represented in Figure 1(b). The

Table 1. Geometric dimensions of the double pendulum setup components.

Parameter	Dimensional Unit
l_1	160 mm
l_2	140 mm
l_3	20 mm
b	48 mm
h	10 mm
m_1	0.555 kg
m_2	0.215 kg

rectangular base of the experimental setup is firmly clamped to a rigid slab. The background of the setup is covered with red coloured cloth for better contrast while tracking. The primary pendulum is white anodized while the secondary pendulum is black anodized to aid in tracking them independently. The orientation of the DSLR camera is calibrated using a tripod stand ensuring that the optical axis passes through the axis of the primary hinge. The focal length of the lens is adjustable in the range 18–55 mm with an aperture range of f/3.5–f/5.6. The motion of the pendulum can be captured within the range of 24–60 fps and the shutter speed can range between 1/30–1/4000 sec as per the camera specifications. The positional data is determined by tracking the pendulums using image processing, the flowchart of which is illustrated in Figure 2.

The input video is in the **.mp4** format. The upper hinge position is manually determined and the image is smoothened by using a bilateral filter so as to retain the edges of the pendulums. Two copies of each frame are considered independently to threshold the pixel values representing the upper and the lower pendulums. The threshold values are manually determined by iteration till the individual frames account

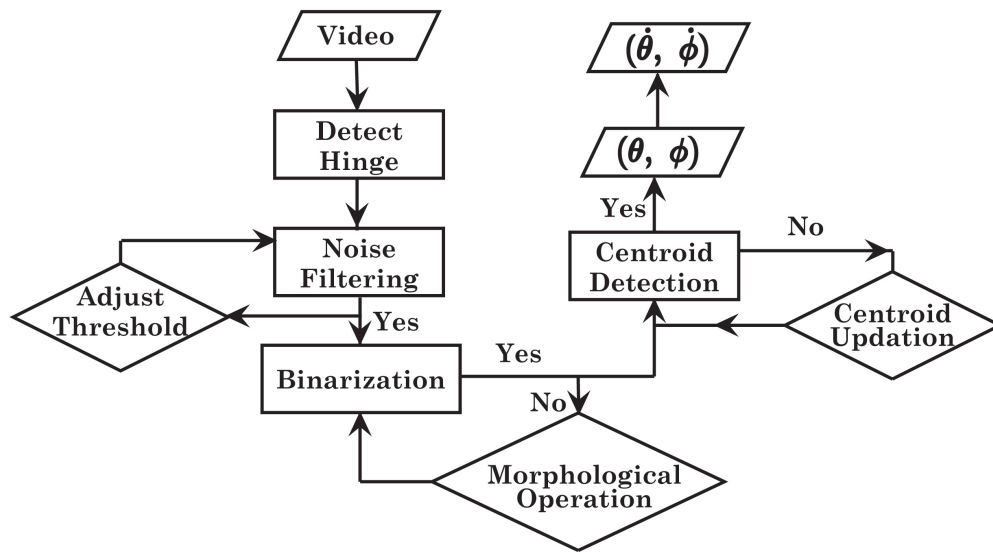


Figure 2. Experimental setup, schematic diagram and fabricated model.

for the majority of the pixels of the respective pendulums. This is followed by binarization to convert the region of the respective pendulums to white pixels on a black background. Unwanted voids and noises are replaced by using morphological operations. These binary images are used to determine the centroid positions of the individual pendulums. When the secondary pendulum is completely eclipsed, the position of its centroid is updated with that of the primary pendulum. The secondary hinge position is determined based on the length of the primary pendulum. The instantaneous angular positions of the individual pendulums are determined by calculating the arctan function of the individual centroid positions with the respective hinge positions. The instantaneous angular velocities of the pendulums in the current frame are determined by numerical differentiation of the angular positions from the neighbouring frames using the formulae, $\dot{\theta}_i = (\theta_{i+1} - \theta_{i-1}) / (t_{i+1} - t_{i-1})$ and $\dot{\phi}_i = (\phi_{i+1} - \phi_{i-1}) / (t_{i+1} - t_{i-1})$, where t is the total time elapsed from the instant of releasing the system till i^{th} frame of the video. The velocity for the boundary cases is considered to be the same as the respective adjacent instantaneous velocities. The complete sequence is scripted using MATLAB software.

3. Governing equations

The distributed mass pendulums are lumped into the model as shown in Figure 3. The angular displacements are θ and ϕ for the upper and lower pendulums respectively. The total kinetic energy (E_K) is given by,

$$E_K = \frac{1}{24} \left[\dot{\theta}^2 (m_1 \alpha + 12m_2 l_1^2) + \dot{\phi}^2 m_2 \beta + \dot{\theta} \dot{\phi} (12\gamma \cos \psi) \right] \quad (1)$$

where, $\alpha = b^2 + 4l_1^2 + 4l_3^2 + 4l_1 l_3$, $\beta = b^2 + 4l_2^2 + 4l_3^2 - 4l_2 l_3$, $\gamma = m_2 l_1 (l_2 - l_3)$ and $\psi = \theta - \phi$. The potential energy (E_P) is given by,

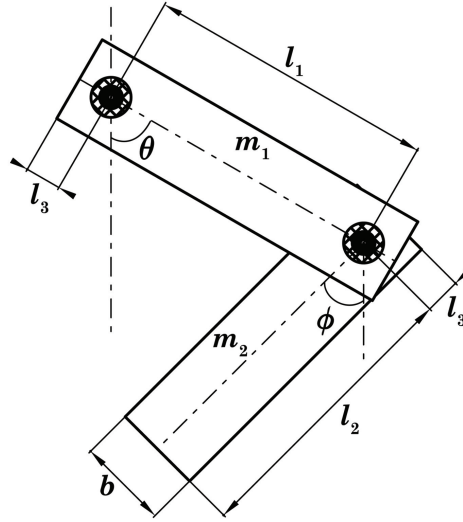


Figure 3. Lumped model of the double pendulum.

$$E_P = \frac{1}{2} \left[(m_2\delta + m_1l_1)g - (m_1 + 2m_2)l_1g \cos \theta - \frac{\gamma}{l_1}g \cos \phi \right] \quad (2)$$

where, $\delta = l_2 - l_3 + 2l_1$. The Lagrangian ($\mathcal{L}$) is defined as $\mathcal{L} = E_K - E_P$, while the energy dissipation due to viscous and friction effects is accounted for by the Rayleigh dissipation function ($\tilde{D}$). The Rayleigh dissipation function is given by $\tilde{D} = \frac{1}{2} \sum_{i=1}^2 \xi_i \dot{\theta}_i^2$, where ξ_i is the dissipation factor. The Rayleigh dissipation function has a unit of energy while the dissipation factor has a unit of $\text{kg-m}^2/\text{s}$. By the inclusion of the non-conservative energy into the Lagrangian formulation, it takes the form,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) = \frac{\partial \mathcal{L}}{\partial \theta_i} - \frac{\partial \tilde{D}}{\partial \dot{\theta}_i} \quad (3)$$

The equations of motion are expressed as follows:

$$\ddot{\theta} = \frac{A_1 + B_1 - C_1}{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2 \cos^2 \psi} \quad (4)$$

where,

$$\begin{aligned} A_1 &= \dot{\theta}^2 (-18\gamma^2 \sin 2\psi) - \dot{\phi}^2 (6m_2\beta\gamma \sin \psi) \\ B_1 &= \left(6\frac{\gamma}{l_1}g \sin \phi + 12\xi_2\dot{\phi} \right) 6\gamma \cos \psi \\ C_1 &= \left[6(m_1 + 2m_2)l_1g \sin \theta + 12\xi_1\dot{\theta} \right] m_2\beta \\ \ddot{\phi} &= \frac{A_2 + B_2 - C_2}{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2 \cos^2 \psi} \end{aligned} \quad (5)$$

where,

$$\begin{aligned} A_2 &= 18\dot{\phi}^2\gamma^2 \sin 2\psi + 6\dot{\theta}^2\gamma \sin \psi (12m_2l_1^2 + m_1\alpha) \\ B_2 &= \left[6(m_1 + 2m_2)l_1g \sin \theta + 12\xi_1\dot{\theta} \right] 6\gamma \cos \psi \\ C_2 &= \left(6\frac{\gamma}{l_1}g \sin \phi + 12\xi_2\dot{\phi} \right) (12m_2l_1^2 + m_1\alpha) \end{aligned}$$

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These equations are solved numerically using the implicit method(s). The dissipation factors ξ_1 and ξ_2 are used as fitting parameters to compare the numerical results with the experimental findings and validate the lumped model.

4. Results

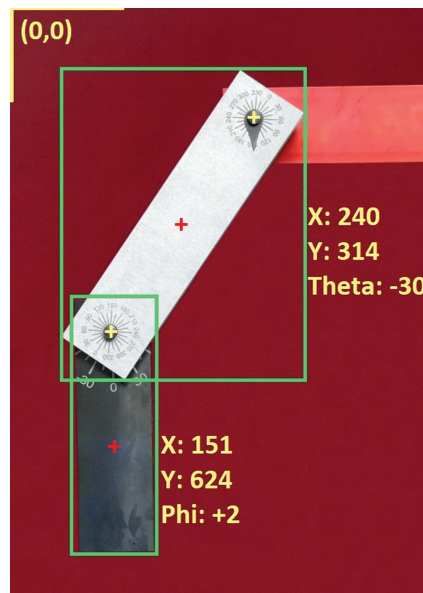


Figure 4. Tracking of the double pendulum with the yellow coloured '+' being the hinge positions and the red coloured '+' being the centroidal positions. The datum for the coordinate measurement is the top left corner of the image.

The experiment is conducted in a closed room with a dry bulb temperature of 30°C, wet bulb temperature of 23°C and relative humidity of 55% which has a considerable effect on the viscosity of the air [40]. The maximum frequency of the system during chaos is measured to be 4.4 Hz, which is greater than the undamped natural frequencies ($\omega_{n1} = 1.152 \text{ Hz}$, $\omega_{n2} = 2.199 \text{ Hz}$) of the system obtained from (A.3). The minimum limit of the frame rate is determined to be 22 fps [41], thus videos are captured at 25 fps for better computational efficiency, along with a shutter speed of 1/800 sec to ensure minimum blurriness [42], which is determined manually to ensure sufficient exposure to sustain better quality of the video.

The angular positions are mapped with the time steps and the video is reconstructed with the bounding boxes along with the coordinates of respective centroids as shown in Figure 4. The corresponding velocities of the pendulums are obtained by finding the spatial derivative from the available positional data. In chaotic motion, the complete interference of the secondary pendulum by the primary pendulum is overcome by replacing the centroid of the secondary pendulum with the primary pendulum. The deviation in the measurement of the position of the secondary pendulum when it is partially interfered is found to be insignificant. There exist other inevitable sources

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of error in the video processing technique, which includes the errors from improper calibration with the reference plane, rolling shutter effect, insufficient resolution, and unstable illumination. These errors are non-additive in nature which makes the cumulative error negligible for all purposes within the scope of this work.

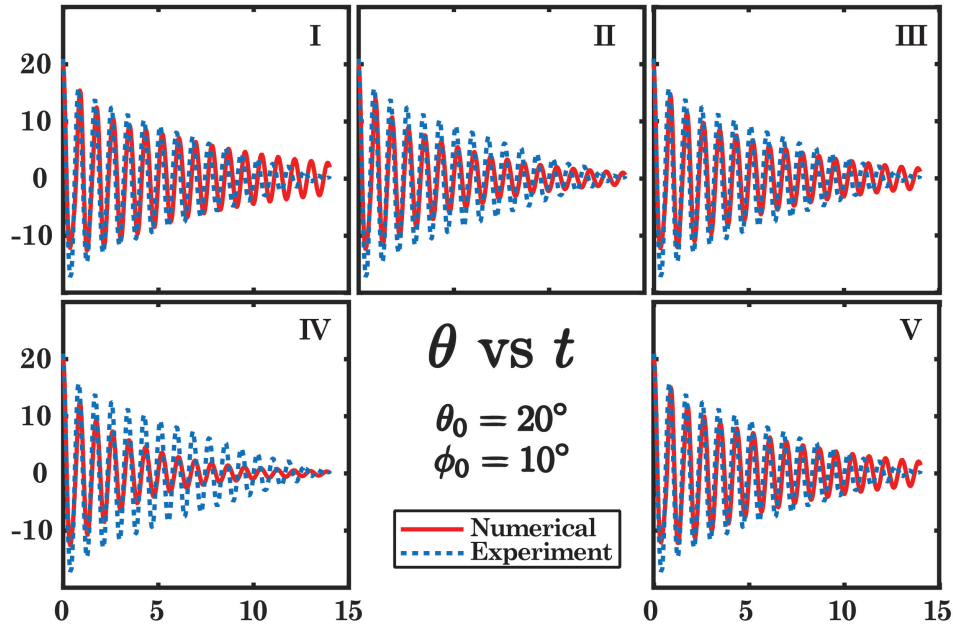


Figure 5. Tuning of the numerical results to fit with the experimental findings by varying the dissipation factors ξ_1 and ξ_2 for initial conditions $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$.

The experimental results of the positional data and the angular velocities of both the pendulums are plotted independently with respect to time with a time step of 1/25 fps, i.e., 0.04s. The optimum values of the dissipation factors are decided by manually tuning the numerical results with respect to the experimental results on the basis of the percentage error in each plot. The average error (ε) is calculated and expressed as a percentage for the angular displacements $\theta(t)$ given by,

$$\varepsilon(\theta) = \frac{1}{n} \sum_{t=0}^{t=t_{\text{end}}} \frac{\theta_{\text{exp}} - \theta_{\text{num}}}{\theta_{\text{exp}}} \quad (6)$$

with a time step of 0.04s, where θ_{exp} and θ_{num} are the experimental and numerical angular displacements of the upper pendulum respectively and t_{end} is the time instant at which the pendulum oscillations become zero after complete die down. Similarly, the average errors $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ corresponding to $\phi(t)$, $\dot{\theta}(t)$ and $\dot{\phi}(t)$ respectively, are determined and presented in Table 2. This procedure is illustrated in Figure 5 for the combination of initial angular displacements $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$. In Case I, the error $\varepsilon(\theta)$ is minimized and is found to be approximately equal to 5% when the dissipation factors are $\xi_1 = 10.507 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 13.493 \times 10^{-4} \text{kg-m}^2/\text{s}$. However, for the same set of dissipation factors the errors $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are found to be 30%, 9%, and 40% respectively as shown in Table 2. Similarly, in Case II, the average error $\varepsilon(\phi)$

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is found to be approximately 7% for the dissipation factors $\xi_1 = 17.910 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 21.254 \times 10^{-4} \text{kg-m}^2/\text{s}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are found to be 11%, 15%, and 20% respectively. For Case III, the average error $\varepsilon(\dot{\theta})$ is found to be approximately 4% for the dissipation factors $\xi_1 = 13.851 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 17.030 \times 10^{-4} \text{kg-m}^2/\text{s}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\phi)$ and $\varepsilon(\dot{\phi})$ are found to be 10%, 22%, and 22% respectively. Eventually, for Case IV, the average error $\varepsilon(\dot{\phi})$ is found to be approximately 10% for the dissipation factors $\xi_1 = 26.269 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 30.209 \times 10^{-4} \text{kg-m}^2/\text{s}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\phi)$ and $\varepsilon(\dot{\theta})$ is found to be 15%, 8%, and 12% respectively.

Table 2. Dissipation factors(in $\text{kg-m}^2/\text{s}$) for varying cases of minimizing the errors(in %) $\varepsilon(\theta)$, $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$.

Case	Dissipation Factor	$\varepsilon(\theta)$	$\varepsilon(\phi)$	$\varepsilon(\dot{\theta})$	$\varepsilon(\dot{\phi})$
I	$\xi_1 = 10.507 \times 10^{-4}$ $\xi_2 = 13.493 \times 10^{-4}$	5	30	9	40
II	$\xi_1 = 17.910 \times 10^{-4}$ $\xi_2 = 21.254 \times 10^{-4}$	11	7	15	20
III	$\xi_1 = 13.851 \times 10^{-4}$ $\xi_2 = 17.030 \times 10^{-4}$	10	22	4	22
IV	$\xi_1 = 26.269 \times 10^{-4}$ $\xi_2 = 30.209 \times 10^{-4}$	15	8	12	10
V	$\xi_1 = 15.000 \times 10^{-4}$ $\xi_2 = 15.000 \times 10^{-4}$	6	10	6	18

From Figure 5, the conclusion can be drawn that the minimization of $\varepsilon(\theta)$ is resulting in large errors of $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ as presented in Table 2. A similar trend is observed in cases II, III and IV, where errors with respect to ϕ , $\dot{\theta}$ and $\dot{\phi}$ respectively are at a minimum. An optimum choice for the dissipation factors are $\xi_1 = 15.000 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 15.000 \times 10^{-4} \text{kg-m}^2/\text{s}$, which are determined by tuning the numerical results and presented as Case V. Although, the errors $\varepsilon(\theta)$, $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are not as low as their individual minimum values in the first four cases, the cumulative error is relatively less, which are found to be 6%, 10%, 6% and 18% respectively. This entire procedure is followed for $0 \leq (\theta_0, \phi_0) \leq 180^\circ$ with an increment of 1° , totalling 32760 experiments. However, this procedure is not entirely for chaotic motion as only one set of dissipation factors are chosen to predict the behaviour. From Table 2 it can be observed that when the error is optimized with respect to $\varepsilon(\theta)$ or $\varepsilon(\dot{\theta})$, the dissipation factors are relatively low when compared to that of the instances of error optimization with respect to $\varepsilon(\phi)$ or $\varepsilon(\dot{\phi})$. Thus an optimum balance of the dissipation factors is established, albeit done by trial and error method.

The results shown in Figure 5 presented the procedure to fit the dissipation factors

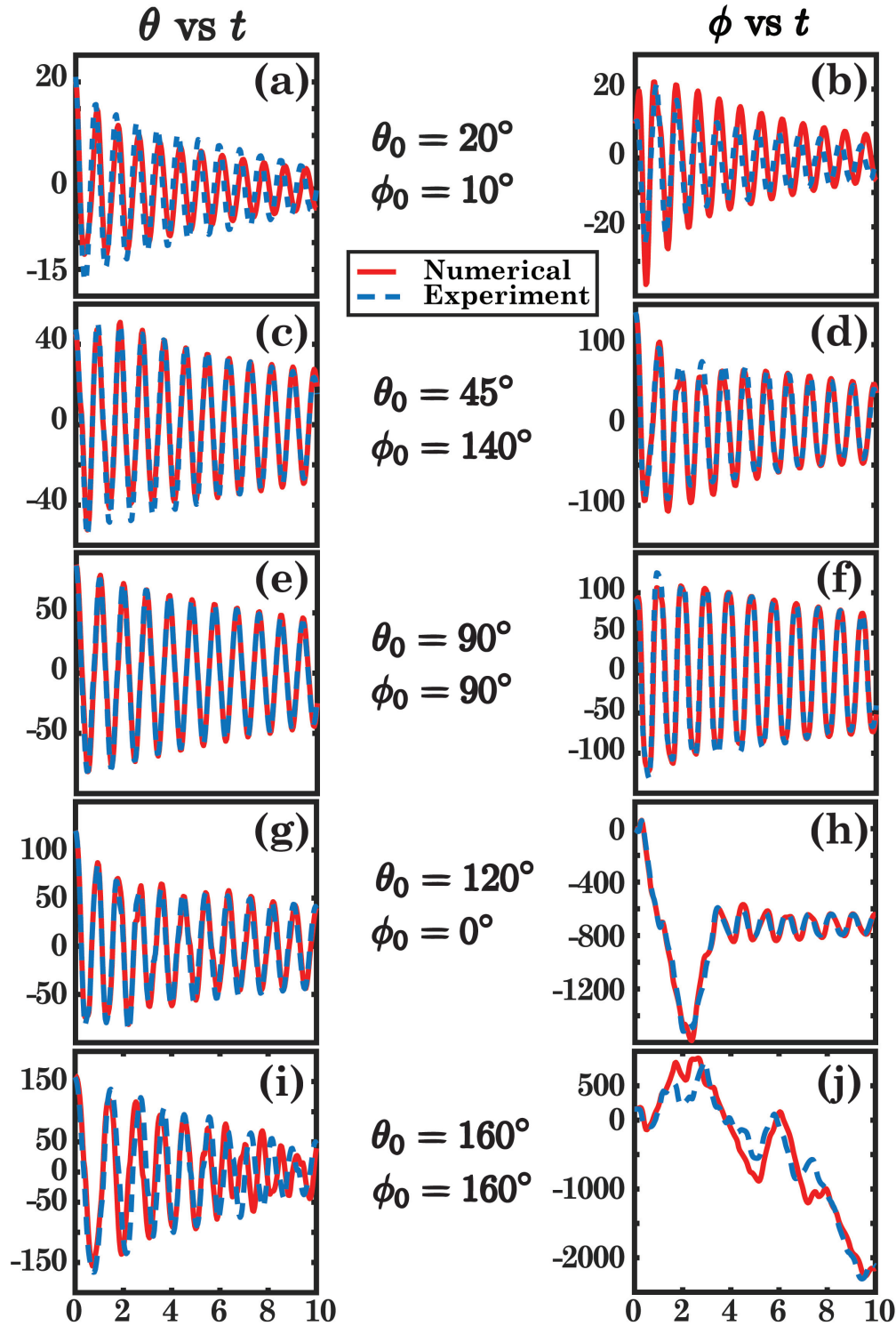


Figure 6. Angular displacements for the primary and the secondary pendulums.

for the initial angular displacements $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$. However, at large angular displacements, it was observed that the dissipation factors that minimize the error during the initial 40% of the total oscillation, resulted in a large error during the small-

angle oscillations. Keeping this in mind, the choice of dissipation factors for large-angle oscillations are based on the experimental findings during the first 40% of the total response. It is observed that for all the instances of chaotic motion, the system is found to display the chaotic motion during the first 10 seconds of its release.

Figure 6 shows the numerical and experimental results during the first 10 seconds of motion for varying initial conditions. Figure 6(a) and Figure 6(b) is for the case of $\theta_0 = 20^\circ$ and $\phi_0 = 10^\circ$ for which the dissipation factors $\xi_1 = 15.0 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 15.0 \times 10^{-4} \text{kg-m}^2/\text{s}$. Figure 6(c) and Figure 6(d) shows the response curves for the initial conditions of $\theta_0 = 45^\circ$ and $\phi_0 = 140^\circ$. The dissipation factors are found to be $\xi_1 = 6.0 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 6.0 \times 10^{-4} \text{kg-m}^2/\text{s}$. Similarly, Figure 6(e) and Figure 6(f) give the responses of the upper and lower pendulums for $\theta_0 = 90^\circ$ and $\phi_0 = 90^\circ$ for which the dissipation factors are $\xi_1 = 4.0 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 6.0 \times 10^{-4} \text{kg-m}^2/\text{s}$. The angular displacements for the initial conditions of $\theta_0 = 45^\circ$, $\phi_0 = 140^\circ$ and $\theta_0 = 90^\circ$, $\phi_0 = 90^\circ$ are obtained numerically without neglecting the nonlinear terms in (4,5).

Figure 6(g) and Figure 6(h) represents the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$ with the dissipation factors $\xi_1 = 3.4 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 5.4 \times 10^{-4} \text{kg-m}^2/\text{s}$. Figure 6(i) and Figure 6(j) represents the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$ with the dissipation factors $\xi_1 = 5.4 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 4.6 \times 10^{-4} \text{kg-m}^2/\text{s}$. The angular displacement ϕ for the initial conditions $\theta_0 = 120^\circ$, $\phi_0 = 0$ and $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$ shows that the lower pendulum exhibits chaotic motion. This procedure is carried out for the entire range of experimental trials and the suitable dissipation factors are determined as well as the respective behavioural regimes are found. It can be seen from Figure 6(g) – (j) that even for chaotic motion, a reasonably good tuning of the dissipation factors is possible. Thus Rayleigh dissipation function can be used as a convenient fitting parameter to assess damping even in a system with strong nonlinearity.

Phase portraits are presented in Figure 7 for the results shown in Figure 6. Figure 7(a) and Figure 7(b) shows the case of $\theta_0 = 20^\circ$ and $\phi_0 = 10^\circ$. Figure 7(c) and Figure 7(d) presents the case of $\theta_0 = 45^\circ$ and $\phi_0 = 140^\circ$. In Figure 7(c) a vertical pinch as θ approaches zero for both positive and negative angular velocity $\dot{\theta}$ can be observed experimentally. Similar behaviour can be observed numerically here for $\xi_1 = 6.0 \times 10^{-4} \text{kg-m}^2/\text{s}$ and $\xi_2 = 6.0 \times 10^{-4} \text{kg-m}^2/\text{s}$ with a reasonably good agreement between the two results. Figure 7(d) shows the phase portrait for the secondary angle ϕ in which a sudden veering of the curve at the right and left extreme ends for a few initial oscillations, which is also captured in the numerical results for the same combination of the dissipation factors.

Figure 7(e) and Figure 7(f) shows the phase portrait for the case of $\theta_0 = 90^\circ$ and $\phi_0 = 90^\circ$. In this case, there is an upward vertical pinch in the trajectory of the phase portrait for the initial few cycles. This is observed when $\dot{\theta}$ is negative. However, in Figure 7(f) there is no sudden veering of the trajectory as observed previously in Figure 7(d). One possible explanation could be that the nonlinearity in the system is a cumulative effect of $\sin \theta$, $\sin \phi$, $\dot{\theta}^2$, $\dot{\phi}^2$, $\sin(\theta - \phi)$ and $\sin 2(\theta - \phi)$ terms as seen in

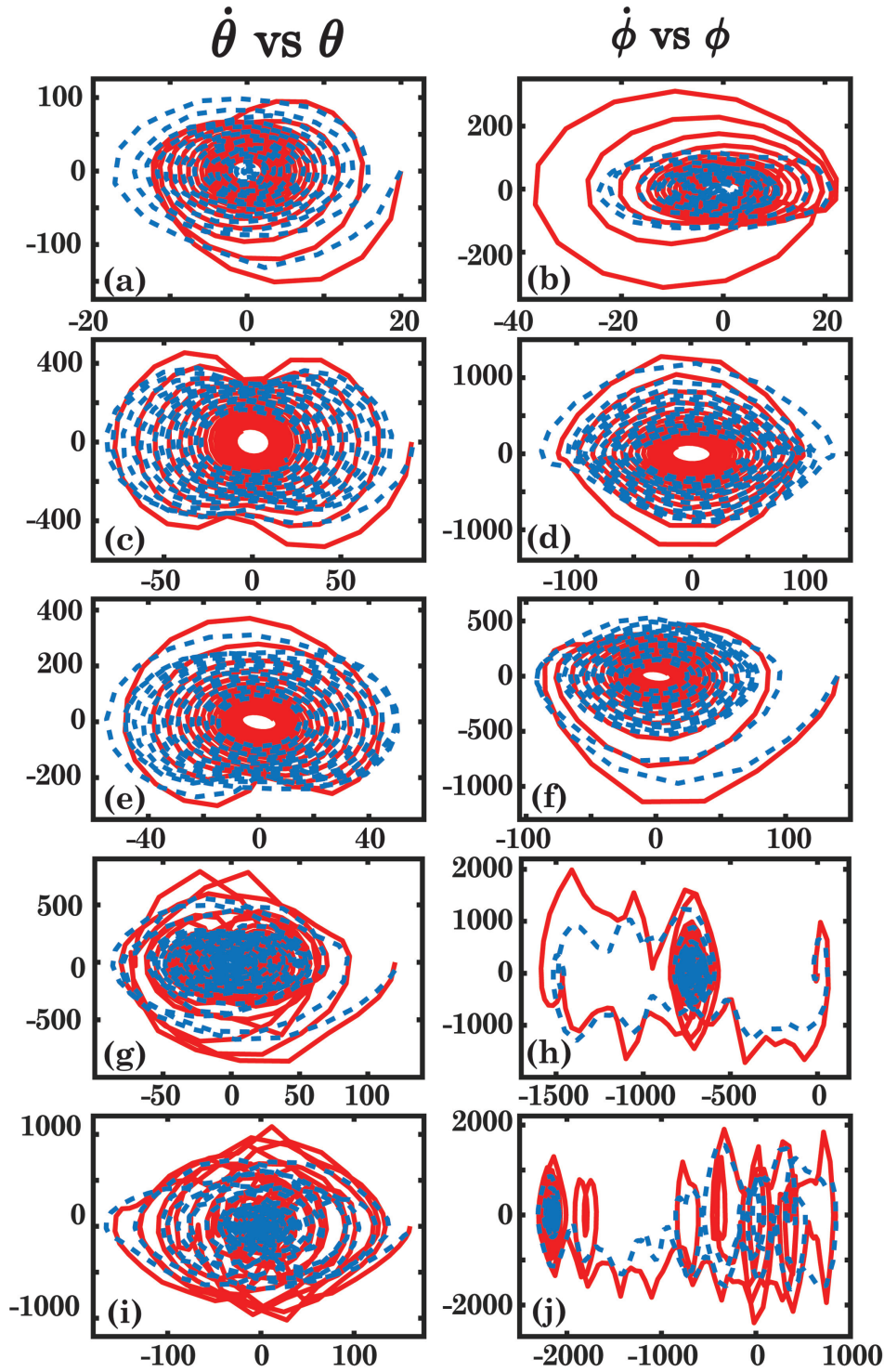


Figure 7. Phase portraits for the primary and the secondary pendulums for the following set of initial angles, (a),(b): $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$; (c),(d): $\theta_0 = 90^\circ$, $\phi_0 = 90^\circ$; (e),(f): $\theta_0 = 45^\circ$, $\phi_0 = 140^\circ$ (g),(h): $\theta_0 = 120^\circ$, $\phi_0 = 0$; (i),(j): $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. The red and blue trend represents the numerical and the experimental results respectively.

(4,5). Hence, for the initial conditions of $\theta_0 = 45^\circ$, $\phi_0 = 140^\circ$ and $\theta_0 = 90^\circ$, $\phi_0 = 90^\circ$ nonlinearity does exist albeit in a dissimilar fashion primarily due to the contribution of ψ and 2ψ , where $\psi = \theta - \phi$.

Figure 7(g) and Figure 7(h) presents the phase portraits for the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$. In Figure 7(h), multiple basins of attraction are visible near the vertical axis passing through $\phi = 0$, -1080° and -1440° before ultimately settling into the basin of attraction where $\dot{\phi} = 0$ and $\phi = -720^\circ$. Figure 7(i) and Figure 7(j) represents the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$. Multiple basins of attraction are numerically observed in Figure 7(j) at $\phi = 720^\circ$, 360° , 0 , -360° , -720° , -1080° , -1800° , whereas experimentally the basins of attraction are observed at $\phi = 720^\circ$, 360° , 0 , -360° , -720° , -1800° . Finally, the lower pendulum settles into the attractor basin at $\phi = -2160^\circ$ as observed experimentally and supported by the simulation results.

The angular displacements of $\theta_0 = 120^\circ$, $\phi_0 = 0$ and $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$ are classified as chaotic motion because of the existence of multiple attractor basins. While fitting the numerical results for the chaotic motion, different basins of attraction are obtained as compared to what is experimentally observed, however, by careful tuning of the dissipation factors, the same basin of attraction is identified, which is found to be unique for a given set of dissipation factors.

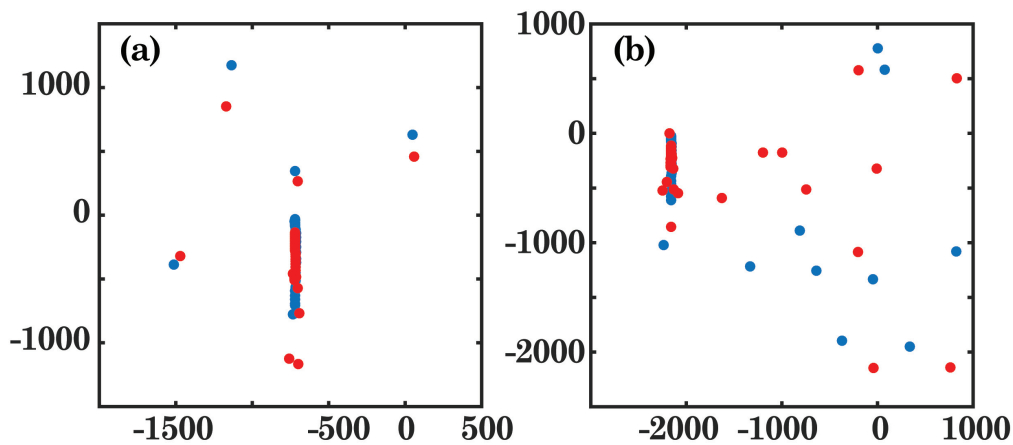


Figure 8. Poincaré maps for the initial angles, (a): $\theta_0 = 120^\circ$, $\phi_0 = 0$; (b): $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. The red and blue dots represents the numerical and the experimental results respectively.

Poincaré sections are used to substantiate the chaotic behaviour of a system. As the phase space of the system of the double pendulum is of four dimensions comprising the variables θ , ϕ , $\dot{\theta}$ and $\dot{\phi}$, the section of the phase space of any two variables forms the Poincaré map when any one of the other two variables is constant. Figure 8 represents the Poincaré maps, which are plotted for the angular velocity of the secondary pendulum against its angular position for all the time instances when the primary pendulum is rotating in a clockwise direction and is at an angular position of 0 degrees. Thus, the maximum number of experimental data points are accounted for.

Figure 8(a) shows the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$ and Figure 8(b) presents the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$. In Figure 8(a) the experimental findings and numerical results; which are tuned by a suitable set of dissipation factors, are found to be collapsing onto the same vertical line passing through $\phi = -720^\circ$. This is also observed in Figure 7(h). Similarly, in Figure 8(b), both the experimental and numerical results are found to be converging to the same basin of an attraction along the vertical line passing through $\phi = -2160^\circ$, which is again observed in Figure 7(j). For a damped oscillation with either periodic or quasi-periodic oscillation, Poincaré maps are found to be having a pattern in their appearance which is observed at the attractor basin as seen in the plots shown in Figures 8(a) and 8(b). However, when the system is exhibiting chaotic behaviour, map points are dispersed randomly all around the phase plane. For the results shown in Figures 8(a) and 8(b), the map points are spread over a wide range of ϕ , which is of the order of $\sim 10\phi_0$. This is used to characterize the chaotic behaviour of the system for the angular displacements of $\theta_0 = 120^\circ$, $\phi_0 = 0$ and $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. Similarly, Poincaré maps are plotted to verify the chaotic behaviour for all the experimental results with varying initial conditions to determine the behavioural regimes — linear, nonlinear and chaotic regimes.

When the average deviation of the simulated results from the results of the linearized equation(A.1,A.2) of motion is less than 1%, the instance of the initial conditions results in linear motion. Linear behaviour is not in any strict sense because of the existence of friction, however, for the sake of brevity in this article, linearity reflects the linear dependency of the system behaviour on θ , ϕ , $\dot{\theta}$ and $\dot{\phi}$. The nonlinearity of the motion is determined for the instances which exhibit the average deviation greater than 1% for the results of the nonlinear equation from the linearized equation. The instances with multiple attractor basins are classified as chaotic behaviour by plotting the phase portraits and hence by extension Poincaré maps.

The behavioural regime of the system is represented in Figure 9. The solid black coloured curve running vertically separates the nonlinear regime on the left from the chaotic regime on the right. The region in between the adjacent red coloured lines represents the region of uncertainty, where, by experiment, the motion is found to be exhibiting both the chaotic and the nonlinear behaviour for the same set of initial angular displacements in multiple trials, which is discussed later. The enclosed region at the origin of the graph represents the linear regime. From the regime, it can be inferred that the chaotic behaviour does exist at relatively lower energy level (eg: $\theta_0 = 90^\circ$, $\phi_0 = 0$) as compared to the nonlinear behaviour that can be observed at a higher energy level (eg: $\theta_0 = 90^\circ$, $\phi_0 = 100^\circ$). This is in agreement with the experimental finding as reported by Himanshu *et al.* [25], in which the dependency of the chaos on the initial energy as well as the initial phase difference has been reported. This phenomenon is observed due to the existence of optimal combinations of the primary and the secondary angles where the lack of energy transfer between the primary and the secondary pendulum helps in sustaining nonlinear behaviour even at higher energy levels.

Region of uncertainty is observed around the boundary separating the chaotic

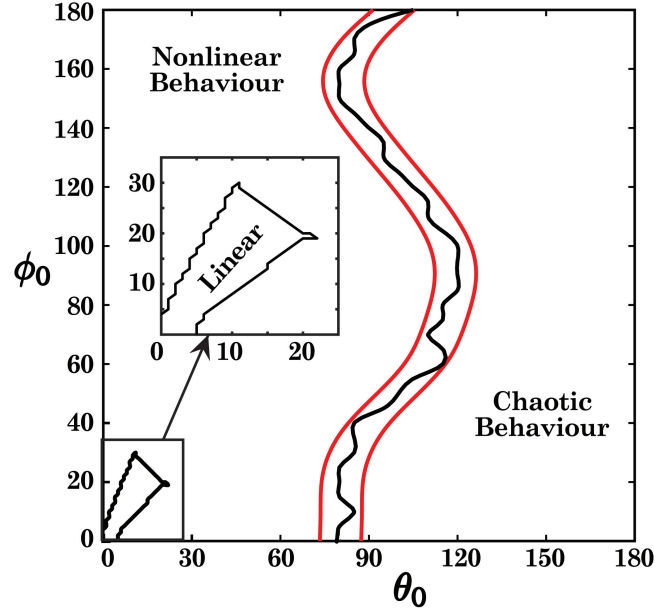


Figure 9. The behavioural regime of the damped autonomous double pendulum factoring range $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$.

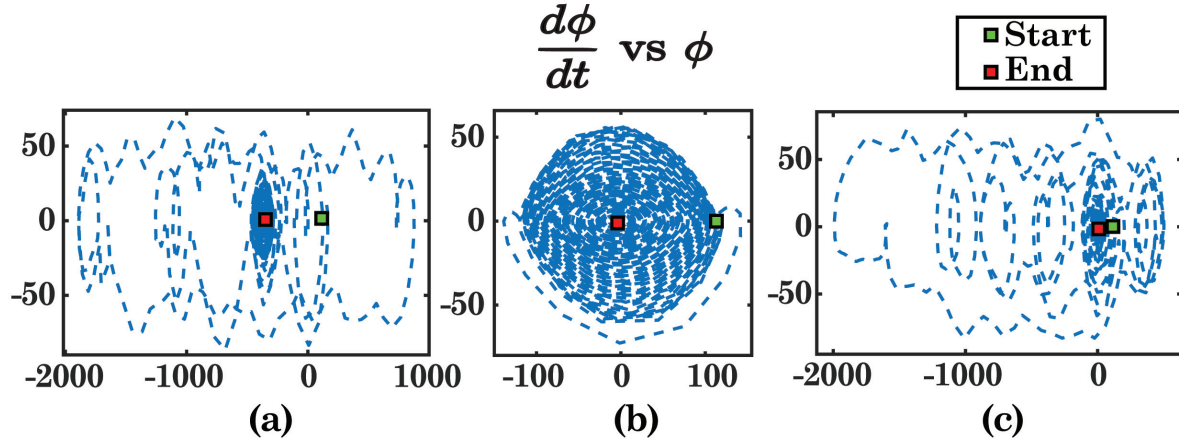


Figure 10. Qualitative uncertainty analysis of the transition between chaotic and nonlinear behaviour for the initial conditions of $\theta_0 = 113^\circ$, $\phi_0 = 117^\circ$.

regime from the nonlinear regime clearly marked by the solid red line in Figure 9. For the same initial conditions, the system is found to be displaying both nonlinear as well as chaotic motion in different trials. This behaviour is augmented by the phase portraits shown in Figure 10. Three cases shown in Figure 10 are for initial conditions of $\theta_0 = 113^\circ \pm 0.1^\circ$, $\phi_0 = 117^\circ \pm 0.1^\circ$. The subplots in Figures 10(a) and 10(c) show chaotic behaviour with different basins of attraction, which is very typical to the chaotic systems. This can be attributed to the fact that identical initial conditions could not be replicated over ~ 50 trials. Therefore, the regime boundary is rather difficult to determine with a high degree of accuracy as compared to the accurate regime boundary, between the

linear and nonlinear regimes.

After analysing the results from Figure 6 and Figure 7, it can be qualitatively concluded that with increasing initial angular displacements, there is a corresponding decrease in the values of the dissipation factors. Hence, as a first-order approximation, the dissipation factor can be written as $\xi_i = \mathcal{F}(\theta_0, \phi_0, \eta, \mu)$, where η is the viscosity of the surrounding medium and μ is the coefficient of friction (rolling or sliding or both). Additionally, since there is damping due to viscous drag, the geometric dimensions also play a significant role in the energy dissipation. Since friction is also involved at the bearing surface, the masses of the pendulums contribute to the energy dissipation as well. The dependency of the dissipation factors ξ_1 and ξ_2 on the initial angular displacements θ_0 and ϕ_0 is depicted in Figure 11.

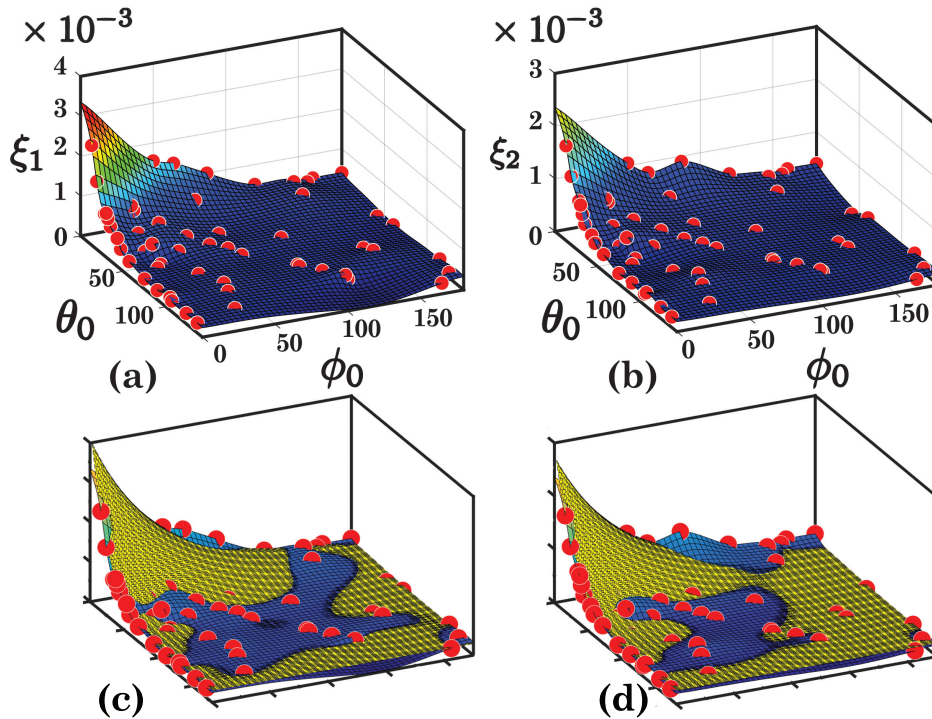


Figure 11. Variation of dissipation factors ξ_1 and ξ_2 obtained from the Rayleigh dissipation function for various initial angular displacements.

Figures 11(a) and 11(b) represents the colour mapped surface fit for the dissipation factors ξ_1 and ξ_2 respectively. The curve-fitting toolbox from MATLAB is used to plot the surface by choosing the 'thinplateinterp' option as fittype. Thick red coloured spheres represent the dissipation factors considered to fit the surface. The dissipation factors presented here are chosen by considering the points having sufficient deviation from the neighbouring ones, such that, the undulations in the surface are clearly captured. From the above plots, it can be stated that the dissipation factors ξ_1 and ξ_2 are relatively low at large and very large initial angular displacements. However, within the limit of small-angle approximation for $\theta \leq 15^\circ$, there is a significant increase in the values of ξ_1 and ξ_2 . This dependency of dissipation factors on the initial angular

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displacements, and hence by extension θ and ϕ has been suggested in the previous section, although in a qualitative capacity. In order to quantify the relationship between dissipation factors and initial angular displacements, a surface equation (7) is fit to the results shown in Figures 11(a) and 11(b).

$$Z_i = \bar{A}_i \cos(\bar{B}_i \theta_0) \cos(\bar{C}_i \phi_0) \exp \left(-\sqrt{\frac{\theta_0^2}{\bar{E}_i} + \frac{\phi_0^2}{\bar{F}_i}} \right) + \bar{D}_i \quad (7)$$

In Figures 11(c) and 11(d) the surface in yellow coloured shade represents the surface fit using equation (7). A total of six arbitrary constants are obtained from equation (7). These constants are determined by minimizing the root mean squared errors for ξ_1 and ξ_2 with respect to Z_1 and Z_2 , given by,

$$\mathcal{R}(\xi_i) = \sqrt{\frac{\sum_{i=1}^n (Z_i(\theta_0, \phi_0) - \xi_i(\theta_0, \phi_0))^2}{n}} \quad (8)$$

where, $\mathcal{R}$ is the root mean squared error, n is the number of data points considered. The minimized root mean squared errors are $\mathcal{R}(\xi_1) = 1.06 \times 10^{-4}$ and $\mathcal{R}(\xi_2) = 0.84 \times 10^{-4}$. The corresponding arbitrary constants in the surface equation(s) are tabulated in Table 3. From Figures 11(c) and 11(d), it can be inferred that both the surface fits are showing a similar trend in the chaotic regime as well as the nonlinear regime. While there is a significant increase in the values of ξ_1 and ξ_2 near the linear regime, this is primarily due to the significant contribution of friction damping. Friction effects are comparatively higher for the upper pendulum because of the higher combined weight acting on its bearing. In the response plots of the nonlinear cases, exponential decay is

Table 3. Arbitrary constants for the best fit surface for the variation of dissipation factors as a function of initial conditions as presented in (7).

Constants	$i = 1$	$i = 2$
$\bar{A}_i$	0.00370	0.00243
$\bar{B}_i$	1.42	1.40
$\bar{C}_i$	0.01	0.04
$\bar{D}_i$	0.00029	0.00025
$\bar{E}_i$	0.126	0.165
$\bar{F}_i$	0.635	0.635

observed until the amplitude reaches the range of $12^\circ - 15^\circ$ and from thereon linear decay is observed till the complete die down of the oscillations, as shown in Figure 12. This phenomenon is due to the dominance of viscous damping in the region of exponential decay that is generally associated with the large amplitudes. At smaller amplitudes rolling friction is playing a major role in the energy dissipation which is reflected by the region of linear decay as seen in Figures 12(a) and 12(b).

The rate of energy dissipation is observed to be higher when viscous effects dominate. This is supported by the nonlinear decay envelope at higher energy bands due to the large angular displacements and angular velocities as seen in Figures 12(c)

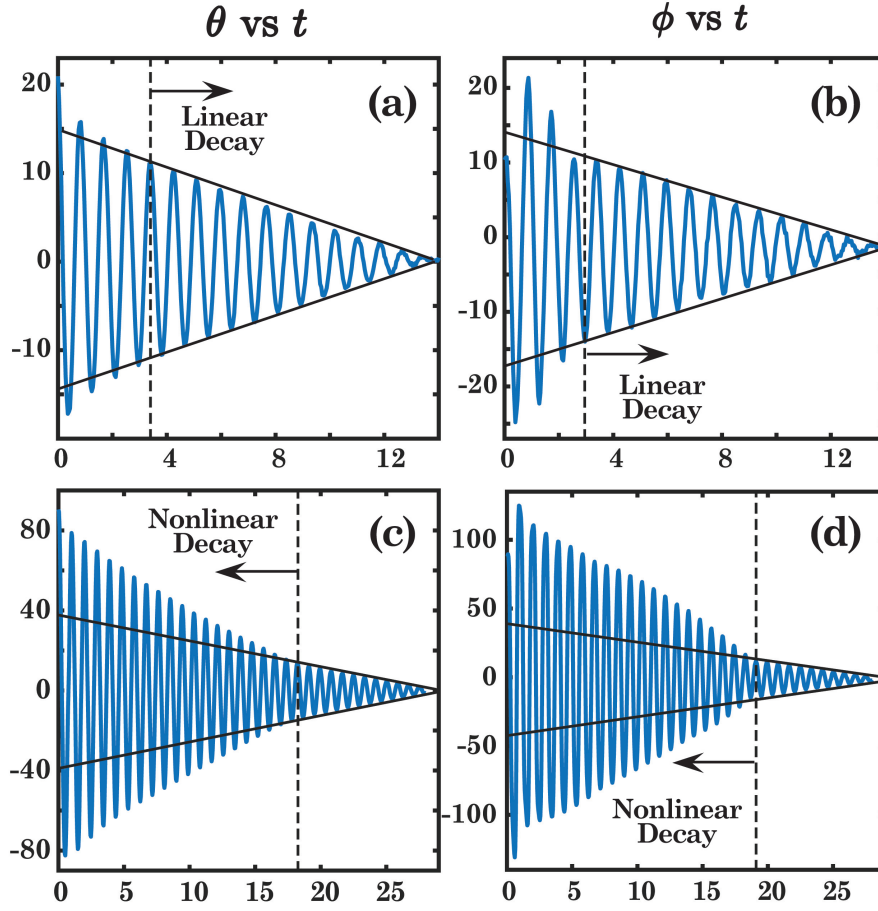


Figure 12. Experimental result of transition from exponential decay to linear decay of the envelope for the initial angles, (a),(b): $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$; (c),(d): $\theta_0 = 90^\circ$, $\phi_0 = 100^\circ$.

and 12(d). However, once a considerable amount of energy has been dissipated, the response curves for both θ and ϕ collapse into the linear decay envelope. The total time required for the oscillations to completely die down is found to reduce with the increase in the amplitude difference ($\psi = \theta - \phi$) between the upper and lower pendulums. As ψ increases, the secondary pendulum has to rotate independently more often about its axis, which leads to greater energy dissipation and thus, the early die down in oscillations. A similar observation has been reported by J.Roy *et al.* [43]. The transition of nonlinear to linear decay suggests the switchover from viscous damping to friction damping is intriguing, to say the least. This interplay of viscous and friction effects on the energy dissipation in pendulum and pendulum systems is an ongoing work.

Experimental studies have been reported for single pendulums [44, 45, 46], where the damping is nonlinearly dependent on the velocity when the viscous effects are dominant and linearly dependent on the velocity when friction effects are dominant. These observations have been experimentally established in this work for a double pendulum system under combined viscous and friction damping effects. Thus, it can be stated that dissipation is also dependent on angular velocities as well and can be

written as,

$$\xi_i = \mathcal{F}(\theta_i, \dot{\theta}_i, \eta, \mu, l_i, b_i, w_i, \rho_i) \quad \text{for } i = 1, \dots, N \quad (9)$$

In the above equation θ_i and $\dot{\theta}_i$ are generalized angular displacements and angular velocities respectively. l_i , b_i , w_i and ρ_i are length, breadth, width and density of the individual pendulum(s). The understanding of the Rayleigh dissipation function and dissipation factor at present is minimal, however, an initial effort to understand its dependency on angular displacements, angular velocities, viscosity effects, friction effects and their influence on the behavioural regimes is carried out in this work and presented in Figure 13.

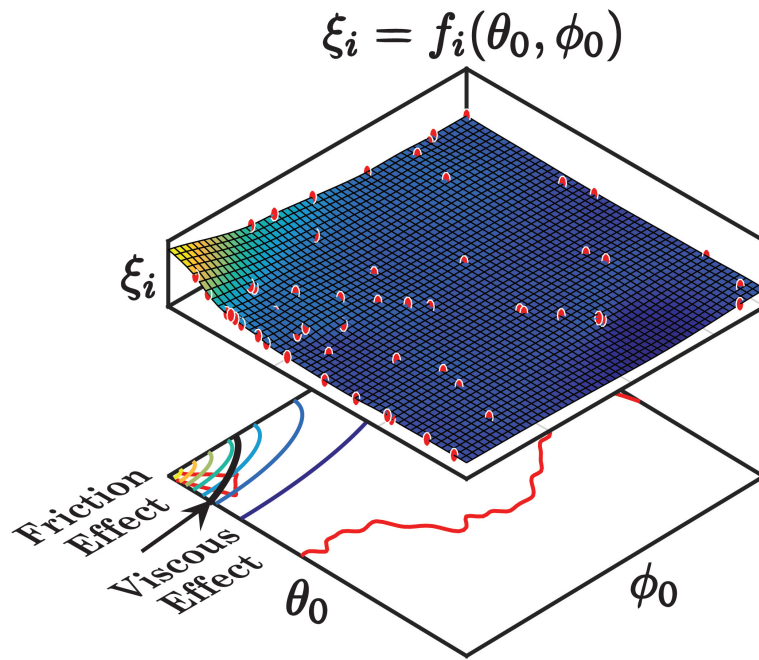


Figure 13. Relation of the plots of behavioural regimes and the surface plot of the Rayleigh dissipation factors.

The onset of linear decay in energy dissipation as seen in Figures 12(a)–(d), which is representative of the damping due to friction, corresponds to a sudden increase in the values of Rayleigh dissipation factors shown in Figure 13. The contours that are shown in Figure 13 are spaced at an interval of $5 \times 10^{-4} \text{kg-m}^2/\text{s}$. Dissipation factor when friction effects are dominant is typically in the range of $1.5 - 2 \times 10^{-4} \text{kg-m}^2/\text{s}$. Since Rayleigh dissipation function is a lumped parameter in the context of this work, quantitative analysis of the effects of viscous and friction damping individually is a work in progress. While the Rayleigh dissipation factor is dependant on angular displacements and angular velocities, the degree and order of its dependence are currently being explored.

5. Conclusions

An experiment setup of a compound double pendulum is fabricated and image-based object tracking is used to measure the angular displacements directly. A lumped parameter mathematical model is derived by including Rayleigh dissipation function to complement the experimental results. Rayleigh dissipation function accounts for the energy dissipation in the double pendulum system subjected to both viscous and friction damping.

The image processing technique is used to track the pendulum which is of non-contact mode and helps to acquire accurate positional data. The linear, the nonlinear and the chaotic regime of the motion are determined experimentally. The experimental data have been analysed for the entire range of combinations of the primary and the secondary angles in the increment of 1° , in the range, $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$. The linear regime of the system is demarcated when the results of the linearized equation(s) are accurate to within 1% of the results obtained from nonlinear equations of motion.

The numerical results are brought to an agreement with the experimental findings by tuning the numerical results by adopting the Rayleigh dissipation factor(s). A surface equation is fit to the Rayleigh dissipation factor(s) to generalize the relationship between the dissipation factors and the initial angular displacements. Determination of the independent contribution(s) of the viscous and friction effects is an ongoing work. The transitional region between the chaotic motion and nonlinear behaviour is presented with the observed uncertainty along the separation boundary.

The dependency of the dissipation factor on the angular velocity of the pendulum needs further examination and is a work in progress. Viscous damping and friction damping are considered to be the major contributors in the dissipation of energy. The viscous effects are significant at large angular amplitudes as compared to the friction effects that are dominant at small angle approximations. The response(s) of the pendulums show a nonlinear decay when viscous damping is prominent, while the decay in the amplitude is linear when friction effects are predominant. The onset of the linear decay after a period of nonlinear decay further establishes that energy dissipation under combined viscous and friction effects can be separately examined based on the varying amplitude scales.

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Appendix-Linearized Equations

After linearization the equation of motion reduces to,

$$\ddot{\theta} = \frac{B_1 - C_1}{\underbrace{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2}_{e_1}} \quad (\text{A.1})$$

where,

$$\begin{aligned} B_1 &= \underbrace{36\gamma^2 \frac{g}{l_1}}_{a_1} \phi + \underbrace{72\xi_2\gamma}_{b_1} \dot{\phi} \\ C_1 &= \underbrace{(6m_1m_2\beta l_1g + 12m_2^2\beta l_1g)}_{c_1} \theta + \underbrace{12m_2\beta\xi_1}_{d_1} \dot{\theta} \\ \ddot{\phi} &= \frac{B_2 - C_2}{\underbrace{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2}_{e_1}} \end{aligned} \quad (\text{A.2})$$

where,

$$\begin{aligned} B_2 &= \underbrace{[36(m_1 + 2m_2)\gamma l_1g]}_{a_2} \theta + \underbrace{72\gamma\xi_1}_{b_2} \dot{\theta} \\ C_2 &= \underbrace{\left(72m_2\gamma l_1g + 6m_1\gamma\alpha \frac{g}{l_1}\right)}_{c_2} \phi + \underbrace{(144m_2l_1^2\xi_2 + 12m_1\alpha\xi_2)}_{d_2} \dot{\phi} \\ \begin{Bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{Bmatrix} &= \frac{1}{e_1} \begin{bmatrix} -c_1 & a_1 \\ a_2 & -c_2 \end{bmatrix} \begin{Bmatrix} \theta \\ \phi \end{Bmatrix} + \frac{1}{e_1} \begin{bmatrix} -d_1 & b_1 \\ b_2 & -d_2 \end{bmatrix} \begin{Bmatrix} \dot{\theta} \\ \dot{\phi} \end{Bmatrix} \end{aligned} \quad (\text{A.3})$$

The undamped natural frequencies obtained from (A.3) are $\omega_{n_1} = 1.152 \text{ Hz}$ and $\omega_{n_2} = 2.199 \text{ Hz}$ and the eigen vectors are $\begin{bmatrix} 0.5525 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -0.2963 \\ 1 \end{bmatrix}$.

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